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Letter to the Editor: Comment on Kerkel et al. (2025) "Does pulse-dose matter? A three-arm comparison of intermittent theta burst stimulation in treatment-resistant depression"

To the Editor,

We read with great interest the recent article by (Kerkel et al., 2025), which investigated the antidepressant effects of repetitive transcranial magnetic stimulation (rTMS), using iTBS (intermittent theta burst stimulation) with varying pulse doses (600, 1200, 1800 per session) in patients with unipolar or bipolar depressive disorder. Their findings revealed no significant differences in treatment outcomes across different pulse counts per session, suggesting that clinical response may depend more on other treatment parameters or individual neurophysiology than on total pulse count per session. In line with these findings, we conducted a retrospective analysis of 94 patients treated at the University Hospital of Psychiatry Zurich, comparing outcomes between those receiving 600 versus 1200 pulses per session. The Swiss Association of Research Ethics Committees granted ethical approval for this analysis under the Nr.: 2025-00559. All patients were diagnosed with a depressive disorder or treatment-resistant depression. Exclusion criteria included acute suicidality, somatic instability, major changes in psychopharmacotherapy or active substance abuse during the rTMS course. Symptom severity was measured via Montgomery–Åsberg Depression Rating Scale (MADRS) and the Beck Depression Inventory-II (BDI-II) scores at baseline and after 19–21 sessions of rTMS. Some participants initially completed the Hamilton Depression Rating Scale (HDRS), which was converted to MADRS scores using established equivalency guidelines (Leucht et al., 2018). Treatment response was defined as a ≥50 % reduction in score from baseline (Reeves et al., 2012; Sforzini et al., 2022). All participants in the 600 pulses-per-session group received (iTBS) targeting the left dorsolateral prefrontal cortex (Lefaucheur et al., 2020). In the 1200 p/s group, treatment protocols included either iTBS over the left dorsolateral prefrontal cortex or 1Hz stimulation over the right, with protocol selection guided by each patient's individual alpha peak frequency (IAPF) (Corlier et al., 2019; Roelofs et al., 2021). Our analysis found that both pulse counts during the treatment protocols led to meaningful symptom improvement, with no significant differences in post-treatment scores (Figs. 1 and 2). Mean (SD) pre- and post-treatment BDI-II scores were 35.32 (14.36) to 28.44 (15.89) in the 600 p/s group and 33.20 (11.23) to 23.35 (13.68) in the 1200 p/s group; MADRS scores decreased from 34.02 (9.33) to 19.86 (8.60) and from 28.78 (8.52) to 17.67 (9.30), respectively. However, a significant group difference was observed in the change in BDI-II scores (ΔBDI), with the 600 p/s group showing a substantially greater reduction compared to the 1200 p/s group ($t(4) = 5.40$, $p < .001$, $d = 1.16$). In contrast, no significant difference emerged for changes in clinician-rated MADRS scores ($t(0) = 1.12$, $p = .264$, $d = 0.23$). Responder rates—defined as a ≥50 % reduction on either scale—did not differ significantly between groups. All non-significant results exceeded the Bonferroni-adjusted threshold of $\alpha = .025$.

These findings align with Kerkel et al.'s core conclusion: clinician-rated outcomes (MADRS) do not significantly differ across stimulation doses. The discrepancy observed in self-reported outcomes—where we found a significant advantage for the 600 p/s group on the BDI-II—may be explained by methodological differences. Most notably, our study did not involve a 1800 p/s condition present in Kerkel et al.'s three-arm design. It is possible that including 1800 p/s diluted potential contrasts between 600 p/s and 1200 p/s in their analysis. Additionally, different self-report instruments were used (BDI-II in our study versus Major Depression Inventory (MDI) in theirs), and our smaller sample size may have introduced sampling variability.

To support direct comparison, we provide a summary table (Table 1) that contrasts key statistical findings across both studies. Overall, while clinician-rated outcomes were consistent, the self-report divergence warrants further exploration—especially as patient-reported outcomes often capture different facets of recovery.

Beyond pulse count, protocol variation within groups may impact results. In our sample, patients receiving 1200 p/s were treated with either iTBS-left or 1Hz-right stimulation. Notably, outcomes within this group did not differ significantly between these subtypes. However, when compared to the uniform 600 p/s TBS-left group, both 1200 p/s subgroups showed reduced self-reported improvement. This pattern suggests that the reduced efficacy in the 1200 p/s condition cannot be explained by protocol heterogeneity alone. Instead, our regression analysis revealed a significant interaction between pulse count and protocol type, indicating that the diminished BDI-II improvement is specific to the 1200 p/s TBS-left configuration. These findings imply that increasing pulse count may not only lack additional benefit, but could, in specific contexts, reduce perceived treatment efficacy.

1. Sensitivity analysis: pulse-dose comparison within iTBS stimulation

To isolate the effect of pulse count from protocol heterogeneity, we conducted a sensitivity analysis restricted to patients who received iTBS. This subgroup included 45 patients treated with 600 p/s and 24 with 1200 p/s. Results confirmed a significant advantage for the 600 p/s group in self-reported symptom change: BDI-II scores showed a substantially larger reduction compared to the 1200 p/s group ($t(2) = 5.13$, $p < .001$, $d = 1.29$), reflecting a large effect size. No significant difference was observed for MADRS change ($t = 0.67$, $p = .505$, $d = 0.17$), and responder rates did not differ between groups on either scale. While post-treatment scores were numerically lower in the 1200 p/s group, between-group differences failed to reach statistical significance (BDI: $p = .072$, $\eta^2 = 0.052$; MADRS: $p = .138$, $\eta^2 = 0.033$). These effect sizes suggest possible trends that may reach significance in larger samples.

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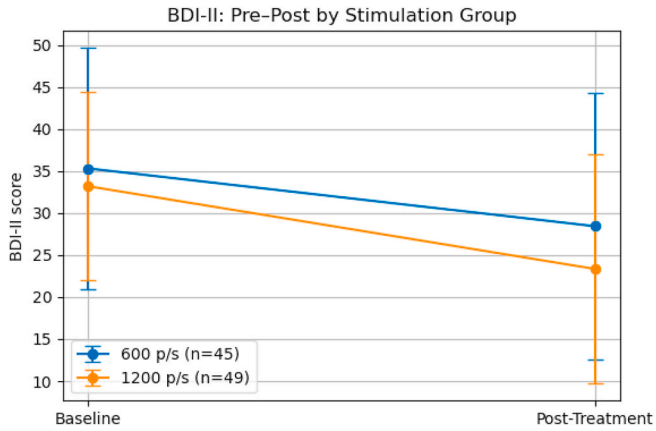

Fig. 1. Pre- and post-treatment self-reported depression severity scores (BDI-II) in patients receiving 600 vs. 1200 pulses per session of rTMS.

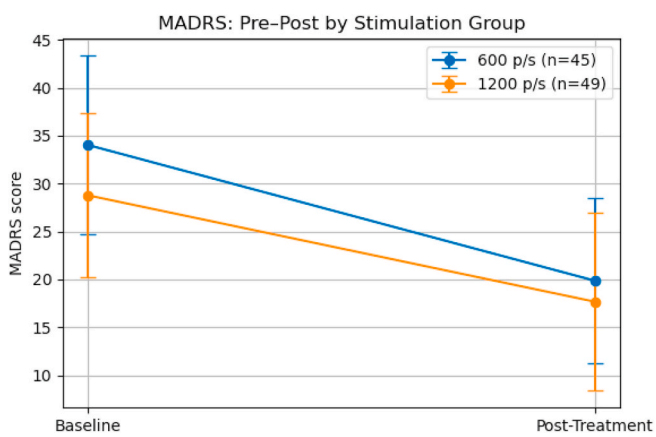

Fig. 2. Pre- and post-treatment clinician-rated depression severity (MADRS) in patients receiving 600 vs. 1200 pulses per session of rTMS.

Overall, despite similar final symptom levels, the 600 p/s group showed significantly greater improvement from baseline, suggesting that pulse count may influence perceived treatment benefit within TBS-left protocols. For example, longer durations of sessions with higher pulse count may have been perceived more stressful in patients, leading to a less pronounced reduction in self-reported symptom severity.

In conclusion, our findings support the main assertion by Kerkel et al. that pulse count per session alone does not determine treatment success in rTMS for depression. However, our data also suggest that, under certain conditions, higher pulse counts may be associated with diminished self-reported benefit. We echo the authors' call for further research to elucidate the neurophysiological and psychological mechanisms underpinning dose-response relationships in rTMS. One question that specifically needs to be addressed is, when and in which patients more stimuli distributed over several sessions per day (accelerated protocols) instead of more stimuili counts per session could boost responses.

CRediT authorship contribution statement

Barbora Provaznikova: Conceptualization, Data curation, Investigation, Writing – original draft. **Samantha Weber:** Data curation, Formal analysis, Methodology, Validation. **Anna Monn:** Supervision. **Cheng Teng Ip:** Supervision. **Erich Seifritz:** Supervision. **Golo Kronenberg:** Supervision, Writing – review & editing. **Sebastian Olbrich:** Conceptualization, Project administration, Supervision, Writing – review & editing.

Table 1
Comparative Outcomes of 600 vs. 1200 Pulses per Session: Divergence from Kerkel et al. (2025) in Self-Reported Depression Improvement.

Item	Kerkel et al. (2025) *(600 vs. 1200 vs. 1800 p/s)*	Current Analysis *(600 vs. 1200 p/s)*	Interpretation/Key Differences
Self-report scale (Kerkel: MDI; ours: BDI-II)	No main effect of group or Time × Group interaction ($F(2, 212) = 1.63$, $p = .198$)	Significant difference: ΔBDI: $t(4) = 5.40$, $p < .001$, $d = 1.16$	Strong self-reported improvement in 600 p/s group; may reflect questionnaire differences (BDI vs. MDI), smaller sample, or absence of 1800 p/s group in current design.
Clinician-rated scale (Kerkel: HAMD-21; ours: MADRS)	No significant group effect ($p = .080$); no Time × Group interaction ($p = .198$)	No significant difference: ΔMADRS: $t(0) = 1.12$, $p = .264$, $d = 0.23$	Results are consistent across studies: clinician-rated outcomes do not vary by pulse count.
Responder rates	Not reported per questionnaire; overall χ^2 not significant	BDI: $\chi^2(1) = 76.0$, $p = .479$ MADRS: $\chi^2(1) = 0.0$, $p = 1.000$	As in Kerkel et al., responder proportions do not differ significantly between groups.
Effect sizes	All partial $\eta^2 \leq 0.028$ (small effects)	BDI: large effect ($d \approx 1.2$) MADRS: small effect	Self-report improvement in 600 p/s group is the only notable divergence in effect size pattern.
Multiplicity control	Bonferroni correction for two scales ($\alpha = .025$)	Identical: Bonferroni $\alpha = .025$	Enables direct comparison of statistical thresholds.

Note. MDI = Major Depression Inventory; BDI-II = Beck Depression Inventory-II; HAMD-21 = Hamilton Depression Rating Scale; MADRS = Montgomery–Åsberg Depression Rating Scale.
Bonferroni-corrected alpha = 0.025 for both studies.

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Declaration of competing interest

All authors declare no conflict of interest.

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